

Ge 158 Lecture 2: Maxwell's Equations and EM Wave Propagation

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Topics covered

- Maxwell's equations
- EM wave propagation in free space
- EM wave propagation in lossy materials
- Poynting vector in general media

1 Review of constitutive parameters

Recall from last lecture that we can describe the EM properties of a material using four parameters, only two of which are relevant for RS:

1. ★ $\epsilon' \epsilon_0$ is the electrical *permittivity* $\rightarrow$ ability to store electric charge;
2. ★ σ is the electrical *conductivity* $\rightarrow$ ability to allow electric current;
3. μ is the magnetic *permeability*; $\mu = \mu_0 = 1/(\epsilon_0 c^2) = 4\pi \times 10^{-7}$ H/m for virtually all RS applications;
4. q_v is the *charge density*; $q_v = 0$ in RS applications.

We group the two important terms into the complex-valued relative dielectric constant

$$\epsilon = \epsilon' - j\epsilon'', \quad \text{where} \quad \epsilon'' = \frac{\sigma}{\omega \epsilon_0}. \quad (1)$$

Here $\sigma/(\omega \epsilon_0)$ is the conduction contribution to ϵ'' ; polarization (relaxation) losses also contribute (Lecture 1). In this lecture, any $\epsilon'' > 0$ (from conduction or otherwise) is what matters.

2 Maxwell's equations

I assume that everyone has some familiarity with classical electrodynamics and the four fundamental equations often called Maxwell's equations. If these are totally new to you, consider consulting any number of excellent texts on the topic: Griffiths, *Introduction to Electrodynamics*; Purcell and Morin, *Electricity and Magnetism*; etc.

2.1 Gauss's law for electricity

Gauss's law for electricity is

$$\nabla \cdot \mathbf{E} = \frac{q_v}{\epsilon' \epsilon_0}, \quad (2)$$

where $\mathbf{E}$ is the electric field intensity vector (units: V/m),

$$\nabla = \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right] = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}, \quad (3)$$

and $\nabla \cdot \mathbf{E}$ is the divergence of $\mathbf{E}$.

2.1.1 Review of divergence

The divergence is a description of how much a vector field spreads out (diverges) at a point (Figure 1a–c).

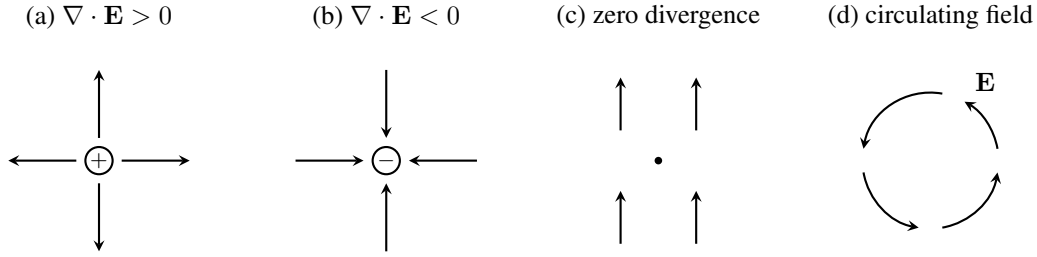


Figure 1: (a) Positive divergence, (b) negative divergence, and (c) zero divergence of a vector field at a point. (d) An example of a divergence-free field that forms closed loops.

- So, in essence, Gauss's law says that a charge causes a flux through some closed surface.
- In other words, like charges repel, and the presence of a charge results in a diverging electric field.
- As noted before, $q_v = 0$ in RS applications $\Rightarrow$ electric fields are divergence-free!
- An example of a divergence-free field is shown in Figure 1d.

2.2 Gauss's law for magnetism

$$\nabla \cdot \mathbf{H} = 0, \quad (4)$$

where $\mathbf{H}$ is the magnetic field intensity vector (units: A/m).

- Fun fact: there is no known magnetic “charge” $\Rightarrow$ magnetic fields are always divergence-free and tend to form closed loops.

2.3 Faraday's law

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}, \quad (5)$$

where $\nabla \times \mathbf{E}$ is the curl of $\mathbf{E}$.

2.3.1 Review of curl

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \hat{\mathbf{z}}. \quad (6)$$

The curl is a measure of the rotation of a vector field about a point.

- Faraday's law tells us that a time-varying magnetic field induces an electric field. (The reverse, a time-varying electric field inducing a magnetic field, comes from Ampère's law with Maxwell's correction; see below.)
- Together with Gauss's law for electricity, it tells us that in RS applications ($q_v = 0$) $\mathbf{E}$ -field lines cannot begin or end on charges, and that a time-varying $\mathbf{H}$ gives $\mathbf{E}$ a non-zero curl (e.g., the closed loops of Figure 1d).

2.4 Ampère's law (with Maxwell's correction)

$$\nabla \times \mathbf{H} = \varepsilon' \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E}. \quad (7)$$

The first term on the right-hand side, $\varepsilon' \varepsilon_0 \partial \mathbf{E} / \partial t$, was Maxwell's correction, and it leads directly to one of the greatest discoveries... as we'll soon see.

- Ampère's law tells us there are two ways to generate a magnetic field:
 1. a time-varying $\mathbf{E}$ field;
 2. an electric current.
 - For $\partial \mathbf{E} / \partial t = 0$, the current $\sigma \mathbf{E}$ generates an $\mathbf{H}$ field that circulates around it (Figure 2a).
 - But when $\partial \mathbf{E} / \partial t \neq 0$, Faraday's law and Ampère's law work together (Figure 2b): a time-varying $\mathbf{H}$ produces a curling $\mathbf{E}$, and a time-varying $\mathbf{E}$ produces a curling $\mathbf{H}$. The result is oscillating $\mathbf{E}$ and $\mathbf{H}$ fields.
- $\Rightarrow$ propagation of a wave $\rightarrow$ it will go on forever unless there is loss ($\varepsilon'' > 0$, e.g., $\sigma > 0$).

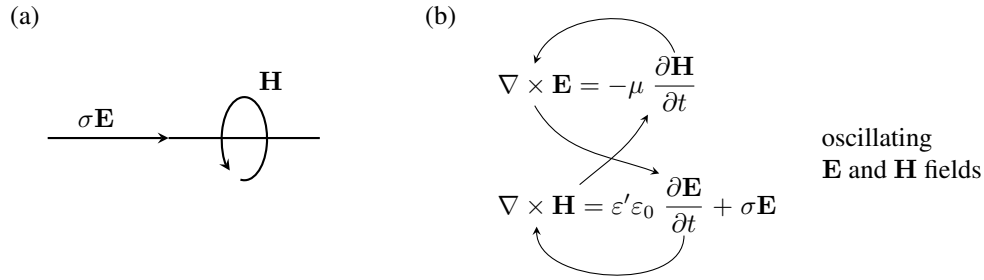


Figure 2: (a) With $\partial \mathbf{E} / \partial t = 0$, the current $\sigma \mathbf{E}$ produces a magnetic field $\mathbf{H}$ that circulates around it. (b) When $\partial \mathbf{E} / \partial t \neq 0$, Faraday's law and Ampère's law feed each other, producing oscillating $\mathbf{E}$ and $\mathbf{H}$ fields.

3 EM wave propagation in free space

3.1 The wave equation

In free space, $\sigma = 0$, $\mu = \mu_0$, $\varepsilon' = 1$, and $q_v = 0$, so Faraday's and Ampère's laws become

$$\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t}, \quad \nabla \times \mathbf{H} = \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (8)$$

For the $\mathbf{E}$ field, take the curl of both sides of Faraday's law:

$$\nabla \times \nabla \times \mathbf{E} = -\mu_0 \nabla \times \left(\frac{\partial \mathbf{H}}{\partial t} \right). \quad (9)$$

Apply the identity

$$\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}, \quad (\nabla^2 = \nabla \cdot \nabla \equiv \text{Laplacian}). \quad (10)$$

Recalling that Gauss's law (here) is $\nabla \cdot \mathbf{E} = 0$,

$$\nabla^2 \mathbf{E} = \mu_0 \frac{\partial}{\partial t} (\nabla \times \mathbf{H}). \quad (11)$$

Plugging in Ampère's law gives the EM wave equation in free space:

$$\nabla^2 \mathbf{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (12)$$

What does this tell us? The wave speed is

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \rightarrow \text{the speed of light.} \quad (13)$$

Because μ_0 and ε_0 are universal constants, c is constant for all EM waves in free space.

For the $\mathbf{H}$ field, we can take the same approach to show

$$\nabla^2 \mathbf{H} = \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}. \quad (14)$$

3.2 Uniform plane waves

Now let's connect these ideas.

- Consider a uniform plane wave propagating along $+\hat{\mathbf{z}}$.
- For convenience, let us define the electric field in terms of a *phasor* (phase vector) that depends on the spatial coordinates x, y, z :

$$\mathbf{E}(x, y, z, t) = \text{Re} \left[\tilde{\mathbf{E}}(x, y, z) e^{j\omega t} \right], \quad (15)$$

where $\tilde{\mathbf{E}}$ is the (complex-valued) phasor, $\text{Re}[\cdot]$ denotes the real component of a complex value, and similarly $\mathbf{H}(x, y, z, t) = \text{Re}[\tilde{\mathbf{H}}(x, y, z) e^{j\omega t}]$ (recall that ω is the angular frequency).

Phasors in brief.

- A quantity oscillating at angular frequency ω , $A(t) = |\tilde{A}| \cos(\omega t + \phi)$, can be written $A(t) = \text{Re}[\tilde{A} e^{j\omega t}]$. The *phasor* $\tilde{A} = |\tilde{A}| e^{j\phi}$ packs the amplitude and phase into one complex number.
- *Why use them:* $\partial/\partial t \rightarrow j\omega$, which turns time derivatives into algebra.
- *Getting back the physical quantity:* multiply by $e^{j\omega t}$ and take the real part. A phasor is not itself the physical field.
- *Products need care:* the time average of a product of two such quantities is $\langle A(t)B(t) \rangle = \frac{1}{2} \text{Re}[\tilde{A}\tilde{B}^*]$. This is the origin of the $\frac{1}{2}$ in the time-averaged Poynting vector, Eq. (44).
- *Conventions differ:* we use $e^{j\omega t}$. Some texts, common in physics, use $e^{-i\omega t}$, which flips the sign of imaginary parts (e.g., $\varepsilon = \varepsilon' + i\varepsilon''$).

- Thus, we have

$$\tilde{\mathbf{E}} = \tilde{E}_x \hat{\mathbf{x}} + \tilde{E}_y \hat{\mathbf{y}} + \tilde{E}_z \hat{\mathbf{z}}. \quad (16)$$

- Substituting the phasor form into the wave equation, Eq. (12), with $\partial/\partial t \rightarrow j\omega$, gives

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k_0^2 \right) \tilde{E}_x = 0, \quad (17)$$

where

$$k_0 = \frac{\omega}{c} = \frac{2\pi}{\lambda_0} \quad (18)$$

is the angular wavenumber in free space and λ_0 is the wavelength in free space.

- A uniform plane wave has no field component along its direction of propagation ($\tilde{E}_z = 0$), and we hereafter define our coordinate system such that $\tilde{E}_y = 0$.
- Because we are considering a uniform plane wave propagating along $\hat{\mathbf{z}}$, we have $\partial^2 \tilde{E}_x / \partial x^2 = \partial^2 \tilde{E}_x / \partial y^2 = 0$, so the wave equation gives

$$\frac{\partial^2 \tilde{E}_x}{\partial z^2} + k_0^2 \tilde{E}_x = 0, \quad (19)$$

for which the solution for a wave propagating along $+\hat{\mathbf{z}}$ is

$$\tilde{E}_x(z) = E_x^+ e^{-jk_0 z} \quad \text{or} \quad \tilde{\mathbf{E}}(z) = E_x^+ e^{-jk_0 z} \hat{\mathbf{x}}. \quad (20)$$

- We can then solve for the corresponding magnetic field by plugging into Faraday's law:

$$\tilde{\mathbf{H}}(z) = \frac{1}{\eta_0} E_x^+ e^{-jk_0 z} \hat{\mathbf{y}}, \quad (21)$$

where

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} \approx 377 \, \Omega \quad (22)$$

is the *intrinsic impedance* of free space.

- For the purposes of comparison, let's rewrite these expressions for $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$:

$$\boxed{\tilde{\mathbf{E}}(z) = E_x^+ e^{-jk_0 z} \hat{\mathbf{x}}, \quad \tilde{\mathbf{H}}(z) = \frac{1}{\eta_0} E_x^+ e^{-jk_0 z} \hat{\mathbf{y}}} \quad (23)$$

3.3 Properties of plane waves

What do Eqs. (23) tell us?

1. They remind us that $\mathbf{E}$ and $\mathbf{H}$ fields are a package deal: you can't have one without the other (although this was clear from Maxwell's equations).
2. They show us that $\mathbf{H} \perp \mathbf{E}$, at least in free space, though this is generalizable to any (isotropic) medium, as we shall see $\Rightarrow$ for plane waves, $\mathbf{H}$ and $\mathbf{E}$ are always orthogonal.
3. $\mathbf{E}$ and $\mathbf{H}$ are in phase, but this is *not* a general property; as we will show, loss (e.g., conductivity) causes a phase shift.
4. $\mathbf{E}$ and $\mathbf{H}$ are related through the intrinsic impedance.

It can be shown that, in 3D,

$$\tilde{\mathbf{E}} = -\eta (\hat{\mathbf{k}} \times \tilde{\mathbf{H}}), \quad \tilde{\mathbf{H}} = \frac{1}{\eta} (\hat{\mathbf{k}} \times \tilde{\mathbf{E}}), \quad (24)$$

where $\hat{\mathbf{k}}$ is the propagation unit vector. This is true for any material with the appropriate definition of η .

The preceding equations make clear that EM (plane) waves are *transverse waves*, meaning they oscillate *only* perpendicular to the direction of propagation (Figure 3). This is a key distinction between EM waves and mechanical waves (e.g., sound waves), which can have longitudinal components.

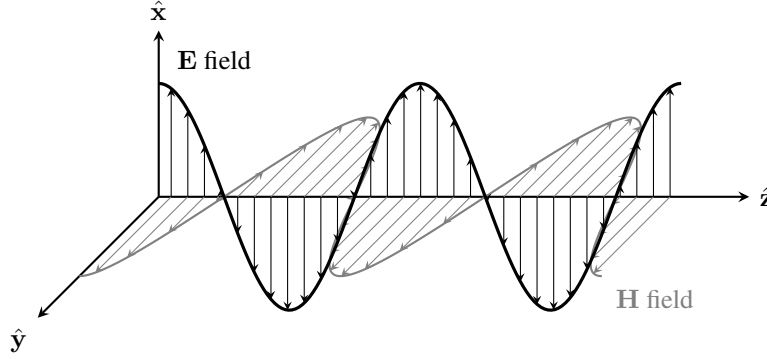


Figure 3: A uniform plane wave in free space propagating along $\hat{\mathbf{z}}$, with $\mathbf{E}$ along $\hat{\mathbf{x}}$ (black) and $\mathbf{H}$ along $\hat{\mathbf{y}}$ (gray), in phase, Eq. (23).

4 EM waves in homogeneous, isotropic, non-ferromagnetic materials

Now we have $\sigma \geq 0$, $\epsilon' \geq 1$, $\mu = \mu_0$, and $q_v = 0$.

- In free space, we saw that all EM waves travel at the same speed and will travel forever (no attenuation).
- What about other materials?

4.1 The wave equation in general media

Recall Ampère's law, Eq. (7), and our phasor form for the fields, Eq. (15). With $e^{j\omega t}$ time dependence, $\partial/\partial t \rightarrow j\omega$, so Ampère's law becomes, in terms of phasors,

$$\begin{aligned} \nabla \times \tilde{\mathbf{H}} &= j\omega\epsilon'\epsilon_0\tilde{\mathbf{E}} + \sigma\tilde{\mathbf{E}} \\ &= j\omega\epsilon_0 \underbrace{\left(\epsilon' - j\frac{\sigma}{\omega\epsilon_0}\right)}_{\text{dielectric constant } \epsilon} \tilde{\mathbf{E}} \\ &= j\omega\epsilon\epsilon_0\tilde{\mathbf{E}}, \end{aligned} \quad (25)$$

and Faraday's law becomes

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu_0\tilde{\mathbf{H}}. \quad (26)$$

Taking the curl of both sides of Faraday's law gives the wave equation in general media,

$$\boxed{\nabla^2 \tilde{\mathbf{E}} - \gamma^2 \tilde{\mathbf{E}} = 0, \quad \gamma = j\omega\sqrt{\mu_0\epsilon\epsilon_0}} \quad (27)$$

where γ is the complex-valued *propagation constant*. We can derive a similar expression for the $\tilde{\mathbf{H}}$ field, $\nabla^2 \tilde{\mathbf{H}} - \gamma^2 \tilde{\mathbf{H}} = 0$.

4.2 Lossless and low-loss media ($\varepsilon \approx \varepsilon'$)

- Define the wavenumber as

$$k \equiv \frac{2\pi}{\lambda} = -j\gamma = \omega \sqrt{\mu_0 \varepsilon' \varepsilon_0} = \frac{\omega}{c} \sqrt{\varepsilon'} = k_0 \sqrt{\varepsilon'}. \quad (28)$$

- Thus, the phase velocity of the EM wave is now

$$u_p = \frac{\omega}{k} = \frac{c}{\sqrt{\varepsilon'}}. \quad (29)$$

It is also common to use $n' = \sqrt{\varepsilon'} = c/u_p$, where n is the index of refraction (Section 4.3.2).

- The phase velocity shows a couple of important points:
 1. $u_p = u_p(\omega)$ when $\varepsilon' = \varepsilon'(\omega) \Rightarrow$ it is common for EM waves to be *dispersive* (waves of different frequencies propagate at different speeds).
 2. $u_p < c$ for $\varepsilon' > 1$.
- Now, following similar steps as before gives the wave equation as

$$\frac{\partial^2 \tilde{E}_x}{\partial z^2} + k^2 \tilde{E}_x = 0, \quad (30)$$

and the solution takes the form

$$\tilde{\mathbf{E}}(z) = E_x^+ e^{-jkz} \hat{\mathbf{x}}, \quad \tilde{\mathbf{H}}(z) = \frac{1}{\eta} E_x^+ e^{-jkz} \hat{\mathbf{y}}, \quad (31)$$

where

$$\eta = \frac{\omega \mu_0}{k} = \sqrt{\frac{\mu_0}{\varepsilon' \varepsilon_0}} = \frac{\eta_0}{\sqrt{\varepsilon'}}. \quad (32)$$

★ Because η is real, $\mathbf{E}$ and $\mathbf{H}$ are in phase.

4.3 Lossy media

Let's add conduction to the mix!

- Recall the wave equations

$$\nabla^2 \tilde{\mathbf{E}} - \gamma^2 \tilde{\mathbf{E}} = 0 \quad \text{and} \quad \nabla^2 \tilde{\mathbf{H}} - \gamma^2 \tilde{\mathbf{H}} = 0, \quad (33)$$

where

$$\gamma^2 = -\omega^2 \mu_0 \varepsilon \varepsilon_0 = -k_0^2 \varepsilon. \quad (34)$$

- For convenience (and convention), let's define

$$\gamma = \alpha + j\beta, \quad (35)$$

where α is the *attenuation constant* and β is the *phase constant*.

- For a uniform plane wave propagating along $\hat{\mathbf{z}}$ with the $\mathbf{E}$ field oriented along $\hat{\mathbf{x}}$,

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} E_x^+ e^{-\gamma z} = \hat{\mathbf{x}} E_x^+ e^{-\alpha z} e^{-j\beta z}, \quad (36)$$

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}} \frac{E_x^+}{\eta} e^{-\alpha z} e^{-j\beta z}, \quad (37)$$

where

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon}} = \frac{\eta_0}{\sqrt{\varepsilon'}} \left(1 - j \frac{\varepsilon''}{\varepsilon'}\right)^{-1/2}. \quad (38)$$

- β acts as the wavenumber.
- α attenuates: the field amplitude decays by $1/e$ over a distance $\delta_s = 1/\alpha$, where δ_s is the *skin depth*.
- The power (intensity) decays by $1/e$ over the *penetration depth* $\delta_p = 1/(2\alpha)$, so $\delta_s = 2\delta_p$.
- Eq. (38) shows us that when $\varepsilon'' > 0$ (e.g., $\sigma > 0$), η is complex-valued $\Rightarrow$ phase shift between the $\mathbf{E}$ and $\mathbf{H}$ fields.

4.3.1 Attenuation and phase constants

Expanding α and β gives

$$\alpha = -\omega \sqrt{\mu_0 \varepsilon_0} \operatorname{Im}(\sqrt{\varepsilon}) = k_0 \left\{ \frac{\varepsilon'}{2} \left[-1 + \sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'} \right)^2} \right] \right\}^{1/2}, \quad (39)$$

$$\beta = \omega \sqrt{\mu_0 \varepsilon_0} \operatorname{Re}(\sqrt{\varepsilon}) = k_0 \left\{ \frac{\varepsilon'}{2} \left[1 + \sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'} \right)^2} \right] \right\}^{1/2}. \quad (40)$$

Sanity checks (compare with the lossless case from before):

- $\alpha = 0$ when $\varepsilon'' = 0$;
- $\beta = k_0 \sqrt{\varepsilon'} = k$ when $\varepsilon'' = 0$.

4.3.2 Connecting to the index of refraction

The index of refraction is commonly used in optics and many other fields. It is defined as

$$n = n' - jn'' = \sqrt{\varepsilon}, \quad n' = \operatorname{Re}(\sqrt{\varepsilon}), \quad n'' = -\operatorname{Im}(\sqrt{\varepsilon}), \quad (41)$$

so that, from Eqs. (39) and (40),

$$\alpha = k_0 n'', \quad \beta = k_0 n'. \quad (42)$$

5 EM power density: the Poynting vector

The flow of power (i.e., energy flux) carried by an EM wave is given by the *Poynting vector*

$$\mathcal{S}(t) = \mathbf{E}(t) \times \mathbf{H}(t) \quad [\text{W/m}^2]. \quad (43)$$

Averaged over time, for a uniform plane wave propagating along $\hat{\mathbf{z}}$ (Eq. (36)),

$$\mathcal{S}_{\text{av}} = \frac{1}{2} \text{Re} [\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*] = \hat{\mathbf{z}} \frac{|E_x^+|^2}{2|\eta|} e^{-2\alpha z} \cos \phi_\eta, \quad (44)$$

where ϕ_η is the phase of η ($\eta = |\eta|e^{j\phi_\eta}$) and $*$ denotes the complex conjugate. Note the factor of 2 in the exponent for power: power decays twice as fast as the field amplitude.

Variables

Units are SI. Values are given for universal constants; “typical” values are representative, not definitive.

Symbol	Description	Units	Value
<i>Universal constants</i>			
ε_0	permittivity of free space	F/m	8.854×10^{-12}
μ_0	permeability of free space	H/m	$4\pi \times 10^{-7} \approx 1.257 \times 10^{-6}$
c	speed of light in a vacuum, $1/\sqrt{\mu_0\varepsilon_0}$	m/s	2.998×10^8
η_0	intrinsic impedance of free space, $\sqrt{\mu_0/\varepsilon_0}$	Ω	≈ 376.7 ($\approx 120\pi$)
j	imaginary unit, $j^2 = -1$ (time dependence $e^{j\omega t}$)	–	–
<i>Constitutive parameters and dielectric properties</i>			
ε	complex dielectric constant (relative), $\varepsilon' - j\varepsilon''$	dimensionless	–
ε'	relative permittivity (real part of ε)	dimensionless	≥ 1 (almost always); free space: 1
ε''	dielectric loss factor; $\sigma/(\omega\varepsilon_0)$ from conduction	dimensionless	free space: 0
μ	magnetic permeability	H/m	$\approx \mu_0$ in RS
q_v	volume charge density	C/m ³	0 in RS
σ	electrical conductivity	S/m	free space: 0
ω	angular frequency of the EM wave	rad/s	–
<i>Fields and vector calculus</i>			
$\mathbf{E}$	electric field intensity vector (instantaneous)	V/m	–
$\tilde{\mathbf{E}}$	phasor of $\mathbf{E}$, $\mathbf{E}(t) = \text{Re}[\tilde{\mathbf{E}}e^{j\omega t}]$	V/m	–
$E_x, \tilde{E}_x$	x -component of $\mathbf{E}$ and of its phasor	V/m	–
$E_y, \tilde{E}_y$	y -component of $\mathbf{E}$ and of its phasor	V/m	0 (chosen)
$E_z, \tilde{E}_z$	z -component of $\mathbf{E}$ and of its phasor	V/m	0 (plane wave along $\hat{\mathbf{z}}$)
$E_x^+, \tilde{E}_x^+$	complex amplitude of the $+z$ -propagating $\tilde{E}_x$ at $z = 0$	V/m	–
$\mathbf{H}$	magnetic field intensity vector (instantaneous)	A/m	–
$\tilde{\mathbf{H}}$	phasor of $\mathbf{H}$	A/m	–
x	Cartesian coordinate	m	–
y	Cartesian coordinate	m	–
z	Cartesian coordinate (direction of propagation)	m	–
$\hat{\mathbf{x}}$	unit vector along x	dimensionless	–

Symbol	Description	Units	Value
$\hat{\mathbf{y}}$	unit vector along y	dimensionless	–
$\hat{\mathbf{z}}$	unit vector along z	dimensionless	–
t	time	s	–
∇	del (gradient) operator	m^{-1}	–
∇^2	Laplacian, $\nabla \cdot \nabla$	m^{-2}	–
<i>Plane-wave propagation</i>			
k_0	angular wavenumber in free space, $\omega/c = 2\pi/\lambda_0$	rad/m	–
λ_0	wavelength in free space	m	–
k	angular wavenumber in a lossless or low-loss medium, $k_0\sqrt{\epsilon'}$	rad/m	–
λ	wavelength in a medium	m	–
$\hat{\mathbf{k}}$	propagation unit vector	dimensionless	–
η	intrinsic impedance of a medium, $\eta_0/\sqrt{\epsilon}$	Ω	real if lossless
ϕ_η	phase angle of η	rad	0 if lossless
u_p	phase velocity, ω/k	m/s	$c/\sqrt{\epsilon'}$ (low-loss)
γ	propagation constant, $\alpha + j\beta$	m^{-1}	$j\omega\sqrt{\mu_0\epsilon\epsilon_0}$
α	attenuation constant	Np/m	0 if lossless
β	phase constant	rad/m	k if lossless
δ_s	skin depth, $1/\alpha$	m	∞ if lossless
δ_p	penetration depth, $1/(2\alpha)$	m	$\delta_s/2$
n	complex index of refraction, $n' - jn'' = \sqrt{\epsilon}$	dimensionless	–
n'	real part of n , $\text{Re}(\sqrt{\epsilon})$	dimensionless	c/u_p ; free space: 1
n''	negative imaginary part of n , $-\text{Im}(\sqrt{\epsilon})$	dimensionless	≥ 0
<i>Power density</i>			
$\mathcal{S}$	Poynting vector (instantaneous power density), $\mathbf{E}(t) \times \mathbf{H}(t)$	W/m^2	–
$\mathcal{S}_{\text{av}}$	time-averaged power density, $\frac{1}{2}\text{Re}[\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*]$	W/m^2	–